Om Bhatt

Engineering Undergraduate (Monash University)
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LinkedIn · **GitHub Portfolio** · **GitHub**

Education

Monash University — Clayton, VIC
Bachelor of Engineering (Honours), Mechatronics — *Expected Grad: 2028*

- **Weighted Average Mark (WAM): 79.300 (Distinction)** · **Grade Point Average (GPA): 3.400**

Technical Skills

- **Hardware Engineering:** Mixed-Signal PCB Design (Altium Designer, KiCad), 4-Layer Stack-Ups & Split-Plane Power, Precision Analog Front-Ends, High-Current Power Distribution, Signal Integrity & EMI Mitigation, LTspice, Soldering.
- **Digital & Embedded:** FPGA Design (Verilog, Quartus Prime, Testbenches), STM32L4 (HAL, DMA, SPI/UART/ADC), ESP32-S3, RP2040, Arduino, C & Assembly.
- **Robotics & Control:** Inverse Kinematics, PID Control Loops, PWM Modulation (18+ channels), Autonomous Navigation, Computer Vision (Object Tracking).
- **CAD & Prototyping:** SolidWorks (Complex Assemblies), Onshape, Rapid Prototyping (FDM), DFM, Linkage & Tendon Mechanism Design.
- **Software & AI:** Python (OpenCV, NumPy, scikit-learn), C++ (OOP, Firmware), Git, ML (Kernel Ridge Regression, CNN Features), MCTS/UCB1, Beam Search, Genetic Algorithms, iOS/Android.

Engineering Experience

Electronics Team Member — *Monash Unmanned Aerial Systems (MUAS)*
Melbourne, VIC — *2026 – Present*

- **Project Cassowary:** Leading hardware development of the Measurement & Logging Board used to characterise heavy-lift drone propulsion systems.
- **Precision Analog Design:** Engineered a 0.1%-precision front-end capturing thrust and torque at **50 kS/s**, using an INA826 instrumentation amplifier to resolve millivolt load-cell signals while rejecting EMI from **120A** motor surges.
- **Design Review:** Validated 40V-rated converters against a 4S LiPo (16.8V peak) worst case for a >2x margin without over-engineering.
- **Mixed-Signal PCB Design:** Routed a 4-layer board in Altium Designer isolating analog front-ends from high-current return paths, with dual-rail LMR14050 buck converters giving the RP2040 logic independent 3.3V/5V domains protected from back-EMF spikes.

Co-Founder — *ARES Technologies (Private R&D Team)*
Melbourne, VIC — *2025 – Present*

- **Industry Recognition:** Selected to exhibit prototypes at the **Monash University DeepNeuron Tech Expo**, presenting alongside industry leaders including Microsoft, Telstra, and Blackmagic Design.
- **Project ECKO (Assistive Haptics):** Designing the 4-layer control board for a haptic-sight blindfold that renders time-of-flight distance as vibration, driving **32 haptic motor channels** from an ESP32-S3 via dual PCA9685 controllers, with a split 3V3/+5V_MOT plane isolating motor noise from the logic and 2.4 GHz radio.
- **Project JAEGAR (Bio-Mimetic Robotics):** Designed an anthropomorphic robotic hand for real-time motion mimicry, engineering a dedicated high-current power bus to eliminate brownouts during simultaneous five-finger actuation and iterating tendon-driven linkages across four SolidWorks generations.
- **Project SCORPION (Autonomous Hexapod):** Built an AI-powered hexapod coordinating **18 servos** on a tripod gait from a custom Python inverse-kinematics solver, replacing failure-prone breadboard wiring with an integrated KiCad PCB that cut chassis weight by **20%**.
- **Project SENTINEL (Computer Vision Defense):** Developed a multi-threaded object tracking system decoupling OpenCV processing from motor control, predicting target paths by vector analysis for real-time turret alignment.

University Projects (Monash)

Warman Design & Build Competition · *Autonomous robot, Arduino Mega* — *MMA2001 Design 1 – 2026*

- Placed **10th of 160 teams**; designed a navigation system that detects deviation and self-corrects mid-run to deliver six payloads within 120 s, keeping the electrical architecture lean for **100% reliability** in final evaluations.

16-bit Multi-Cycle Processor · *Verilog, Quartus Prime, DE10-Lite FPGA* — *ECE2072 Digital Systems – 2026*

- Built the control unit, one-hot tick FSM and parametric registers of an 8-instruction processor with ROM and branching; self-checking testbenches passed **40,323 checks** with zero failures (849 LEs, timing met at 50 MHz).

Real-Time Audio Acquisition System · *C, STM32L432, Python* — *ECE2071 Systems Programming – 2026*

- Dual-STM32 pipeline sampling audio at **44.1 kS/s** via DMA over SPI/UART to a PC CLI, with sample packing, CRC-8 checking and distance-triggered recording; scored **39.75/40** in the demo and viva.

Regulated 3.3V Power Supply Tender · *LM317, LTspice, soldered prototype* — *ECE2131 Electrical Circuits – 2026*

- Supply for an STM32 audio system delivering >140 mA with reverse-polarity and ∼30 V overvoltage protection, costed at **$2.27/unit** at 10k volume (vs $5 target); tender scored **94.7%**.

Software & AI Projects

AI Agent Developer · *Pokemon TCG AI Battle Challenge (Kaggle)* — *Team Project – 2026*

- Deck-building agent over **18,000+ cards** and **250,000+ self-play battles**, using MCTS (UCB1) and beam search, with a genetic algorithm evolving ∼20 evaluation weights.

Founder, Endeavour · *Gamified habit-tracking app (iOS / Android)* — *2026 – Present*